

# OS2 Rationale

## Robustness:

In my developed web application, we can see robustness explicitly in my password hashing for both customers and admins.

```
from werkzeug.security import generate_password_hash, check_password_hash
```

```
hashed_password = generate_password_hash(password)
```

Simply generated bcrypted hashed passwording the prevent the damage caused in a potential data breach securing unwanted account access (thus adhering to the Data protection Act & GDPR). This was promised in the proposal and successfully implemented by us.

### Risk

#### Risk Mitigation

| Risks       | Meaning                                                                     | Consequences                                                         | Severity | Solution                                                   |
|-------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------|----------|------------------------------------------------------------|
| Data Breach | Case of computer misuse (Clear obstruction of the Computer Misuse act 1990) | The release of sensitive user data~ possibly leading to lawsuits etc | HIGH     | Bcrypt hashing to prevent the visibility of sensitive data |

Another clear example of this webapp being robustness is our multiple uses of redirecting, when a non-admin user tries to gain access to our admin dashboard, they are swiftly giving an error message signally firmly they don't have access to the admin dashboard

```
else:  
    # Returns a simple message and the 403 (Forbidden)  
    return "Access Denied! Admins Only. If you have admin privileges Please Log In", 403
```

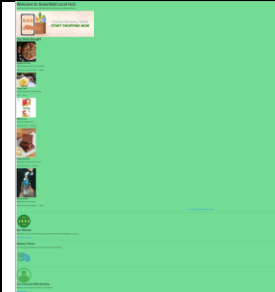
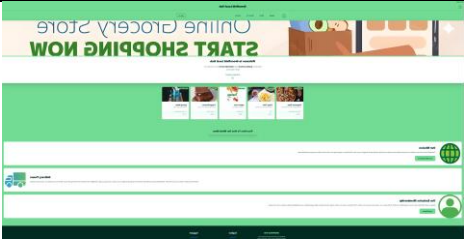
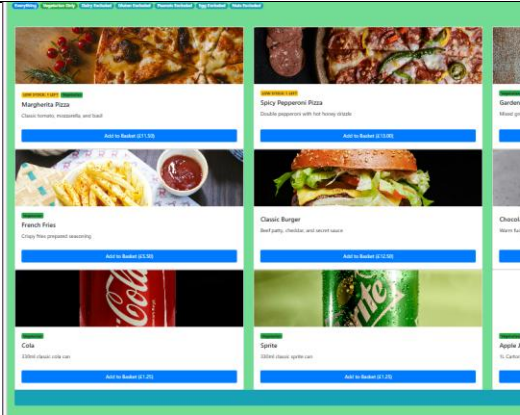
## Code Maintainability:

Our code maintainability can fully be seen in our test log where we didn't just test Normal/Boundary/Erroneous Data; we mostly tested Black/White Box signifying the journey we went to manifest this code, not only our successes but our failures also. Focusing on Black/White Box give the reviewer a greater insight of the code because obviously they see both the troubles/success both backend & frontend deepening their understanding of our developed code

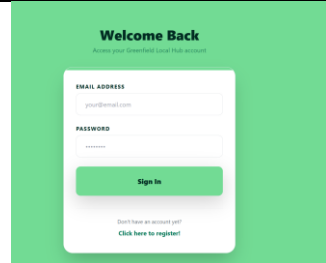
## Evidence/Versioning

Throughout developing with web app, I knew my strengths and weaknesses. In making a website 2 things are paramount! Does it work? How does it look? Making sure that its function of our system was my personal priority because they focus on a website that sells is that everything is working to a

tee without problems. In development sacrifices need to be made so I decided to use tailwinds CSS & bootstrap to increase user satisfaction overall in my styling: here are my changes/evidence:

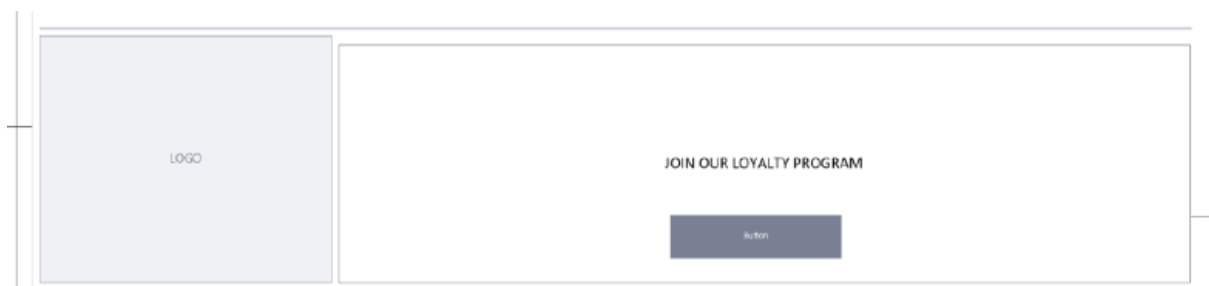
| prototype                                                                                                                                                                                                                                                                                                                                                                                   | Final                                                                               | Reasoning/Indication                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                            |   | Without the CSS Tailwind styling help the web app would have looked tacky and                                                                                                                                                               |
| <div>Margherita Pizza: £11.50<br/>Spicy Pepperoni Pizza: £13.00<br/>Garden Salad: £8.50<br/>Garlic Prawn Pasta: £18.00<br/>Salty Chicken Skewers: £7.50<br/>French Fries: £5.50<br/>Classic Burger: £12.50<br/>Chocolate Brownie: £6.50<br/>Almond Croissant: £3.50<br/>Water: £0.70<br/>Cola: £1.25<br/>Sprite: £1.25<br/>Apple Juice: £2.00<br/>Orange Juice: £2.00<br/>Milk: £2.90</div> |  | Menu page was fully done by me using no external help, simply utilising the bootstrap card template and using my knowledge of python to manifest a fully functional menu page with minimalist design. My prototype was simply used as basis |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                       |
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| <div><p><b>The Vision</b></p><p>The story of Greenfield Local Hub began in the fertile soil of a multi-generational family farm. For the founder, the greatest reward of the harvest was never the profit, but the simple act of providing for those nearby. The farm's true purpose in the early mornings spent tending rows of fresh greens to neighbours, driven by a deep-seated belief that community is only as strong as the hand that nurtures it. However, the years witnessed a growing barrier between the field and the food shelves the land to serve. Traditional supply chains were becoming increasingly complex, leaving growers to use white kitchens tucked in transit. Unwilling to see the fruits of their labour separated from the community by layers of corporate intermediaries, he decided to take a radical step. By combining his lifelong passion for local agriculture with a modern vision for direct-to-door logistics, he founded <b>Greenfield Local Hub</b>.</p><p><b>Early Days</b></p><p>While the foundations were laid in the soil of a single farm, the true evolution of Greenfield Local Hub began when other like-minded producers saw the impact of this direct-to-door mission. Recognising that the community craved more than just fresh produce, the founder began partnering with local artisans, dairy farmers, and specialist growers who shared his uncompromising standards for quality.</p><p><b>Local Heroes</b></p><p>These producers were the catalysts that elevated GLH from a local delivery service into a comprehensive community marketplace. From the baker who provides hand-crafted loaves of bread to the small-scale dairy that ensures every bite is fresh, these partners brought a new level of variety and quality to the region. By uniting under a single banner, these local heroes transformed Greenfield Local Hub into a powerhouse of regional commerce.</p><p><b>Heritage</b></p><p>Today, the heart of our online application marks the beginning of a new chapter for Greenfield Local Hub. What started in the fields is now rising into the digital age, creating a seamless bridge between the heritage of the land and the convenience of modern life. This platform is more than just a store; it's a digital marketplace designed to make supporting local producers easier and more rewarding than ever before. Looking toward the future, GLH remains a place where technology and tradition coexist to strengthen our community even further. While we continue to expand our reach and refine our services, our mission remains fixed: to use this new digital frontier to ensure that the food that grows and doesn't just sit on a shelf, is in your grasp, we remain committed to the soil that nurtured us, using this innovation to protect the future of local farming for generations to come.</p></div> | <div></div> <div><h3>Current</h3><p>While the foundations were laid in the soil of a single farm, Greenfield Local Hub began when other like-minded producers saw the impact of this direct-to-door mission. Recognising that the community craved more than just fresh produce, the founder began partnering with local artisans, dairy farmers, and specialist growers who shared his uncompromising standards for quality.</p><p>These producers were the catalysts that elevated GLH from a local delivery service into a comprehensive community marketplace. From the baker who provides hand-crafted loaves of bread to the small-scale dairy that ensures every bite is fresh, these partners brought a new level of variety and quality to the region. By uniting under a single banner, these local heroes transformed Greenfield Local Hub into a powerhouse of regional commerce.</p></div> <div><h3>Aspirations</h3></div> | Same as home page, getting help with tailwind's style made me about us easier to the eye. I'm sure customers would be satisfied without final product |
| <div><p>Send Email Your Message</p><p>Message</p><p>Email and Phone Number</p><p>Send</p></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <div><h3>Get in Touch</h3><p>Have a question? We'd love to hear from you.</p><div><p>PLEASE ENTER YOUR MESSAGE</p><p>Hello</p><p>EMAIL ADDRESS</p><p>email@example.com</p><p>PHONE NUMBER</p><p>2131</p><p>Send Message</p></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Another use of our tailwind's upgrade to make our front end my appealing to customers. Prototype was                                                  |

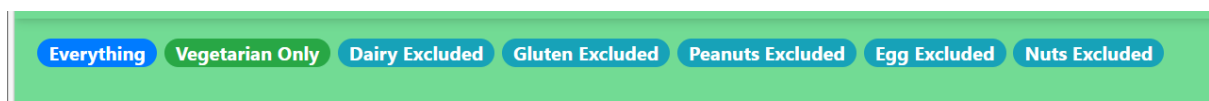
|                                                                                                                                    |                                                                                    |                                                             |
|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------|
|                                                                                                                                    |                                                                                    | functionally but I wanted the front end to look good Aswell |
| <div>Email address</div> <div>Password</div> <div>Submit <a href="#">Don't have an account? Click this link to register!</a></div> |  | Prototype fully functional. Quick talwinds .css upgrade     |

## Comparison/Proposal Changes/Future Changes

I tried basing my focal pages directly from the wireframes I designed from OS1, they mostly adhere to them with a few changes along the way, a great example of this is my missing loyalty program promotion banner that I thought would be of great importance:



Direct comparison wise I mostly adhered to my proposal & design docs; we can see this in implementations like my safety module focusing on allergy responsibility on our part. This was well implemented in our menu page:



### Safety

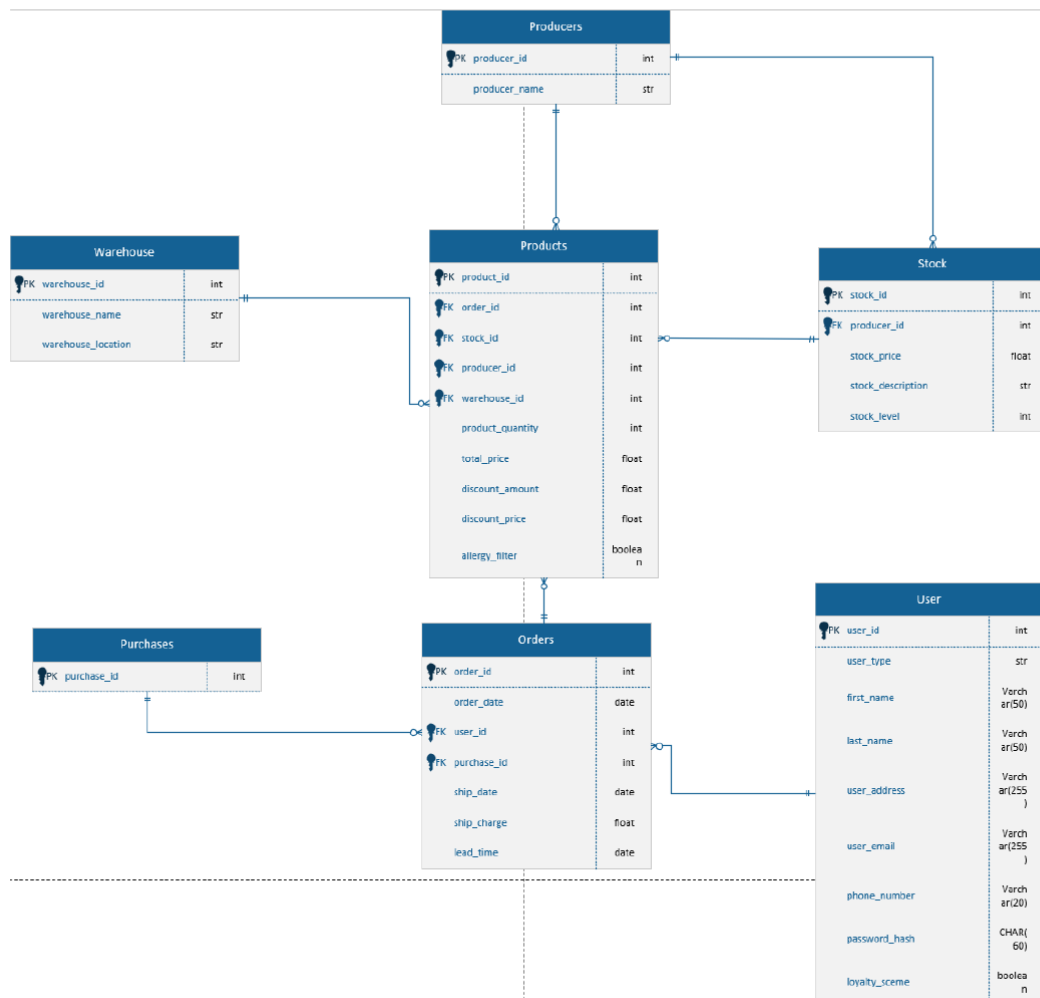
The Safety Module, this module is regarding solely the safety aspects of our product's- importantly labelling/filtering products for user needs:

- Allergies

Our only component is allergies, allergies are DETREMENTAL on every food selling web app since misinformation or slothfulness could cause actual user harm, even death in some cases. This is why I propose we include an allergy filter in the menu page so users can simply pick out their allergy and filter them out. To achieve this, I used a variable ("allergy\_filter") to filter out key allergies ; if a user has a specific allergy not included in the filter, we will strongly advise them to look deeply in the specific products information text. This is our main adherence with the Food Information Regulation.

Significant changes can be seen in my database where I appropriately change sum columns compared to my ERD & Data Dictionary(

## ERD (One too Many)



), as I went on developing this web app I realised certain aspects of my ERD didn't make much logical sense and I made the executive decision to go off track to make my data base fully logical. This doesn't mean I didn't take much inspiration of my design docs since they gave me a basis on what I was trying to grasp. Critical thinking made me make my final choices of what my database should be:

```

class User(db.Model):
    user_id = db.Column(db.Integer, primary_key=True)
    user_type = db.Column(db.String(20), default="customer")
    first_name = db.Column(db.String(50))
    last_name = db.Column(db.String(50))
    user_address = db.Column(db.String(255))
    user_email = db.Column(db.String(255), unique=True, nullable=False)
    phone_number = db.Column(db.String(20), unique=True)
    password_hash = db.Column(db.String(128), nullable=False)
    loyalty_scheme = db.Column(db.Boolean(), default=False, nullable=False)

    # Relationship to find all orders for a user
    orders = db.relationship('Order', backref='customer', lazy=True)

# --- 2. ORDER TABLE (The "Receipt" Header) --- Including The Order primary key(order_id), foreign key(user_id) and attributes.
class Order(db.Model):
    order_id = db.Column(db.Integer, primary_key=True)
    # MAYBE order_date = db.Column(db.DateTime, default=datetime.utcnow, nullable=False)
    order_date = db.Column(db.Date, nullable=False)
    ship_date = db.Column(db.Date)
    ship_charge = db.Column(db.Numeric(10, 2), default=0.00)

    # FK: Link to the User who placed the order
    user_id = db.Column(db.Integer, db.ForeignKey('user.user_id'), nullable=True)

    # Relationship to find all items inside this specific order
    items = db.relationship('Product', backref='parent_order', lazy=True)

# --- 3. STOCK TABLE (The "Master Menu") --- Including The Stock primary key(stock_id), foreign key(warehouse_id & producer_id) and attributes.
class Stock(db.Model):
    __tablename__ = 'stock'
    stock_id = db.Column(db.Integer, primary_key=True)
    stock_name = db.Column(db.String(100), nullable=False)
    stock_price = db.Column(db.Numeric(10, 2), nullable=False)
    stock_description = db.Column(db.Text)
    stock_level = db.Column(db.Integer, default=0)
    is_vegetarian = db.Column(db.Boolean(), default=False, nullable=False)
    allergy_tag = db.Column(db.String(50), nullable=True)

    # FK: Where the stock is made and who made the stock
    warehouse_id = db.Column(db.Integer, db.ForeignKey('warehouse.warehouse_id'))
    producer_id = db.Column(db.Integer, db.ForeignKey('producer.producer_id'))

# --- 4. PRODUCT TABLE (The "Order Line Items") --- Including The Product primary key(product_id), foreign key(order_id & stock_id) and attributes.
class Product(db.Model):
    product_id = db.Column(db.Integer, primary_key=True)

    # FK: Link to the specific Menu Item (Stock) and the Order (Receipt)
    stock_id = db.Column(db.Integer, db.ForeignKey('stock.stock_id'), nullable=False)
    order_id = db.Column(db.Integer, db.ForeignKey('order.order_id'), nullable=False)

    product_quantity = db.Column(db.Integer, nullable=False, default=1)

    # Pricing logic
    total_price = db.Column(db.Numeric(10, 2)) # qty * stock_price
    discount_amount = db.Column(db.Numeric(10, 2), default=0.00)
    discount_price = db.Column(db.Numeric(10, 2)) # total_price - discount_amount

# --- 5. WAREHOUSE & PRODUCERS (Supply Chain) --- Including The Warehouse & Producer primary key(warehouse_id & producer_id) and attributes.
class Warehouse(db.Model):
    warehouse_id = db.Column(db.Integer, primary_key=True)
    warehouse_name = db.Column(db.String(100), nullable=False)
    warehouse_location = db.Column(db.String(255), nullable=False)
    stocks = db.relationship('Stock', backref='location', lazy=True)

class Producer(db.Model):
    producer_id = db.Column(db.Integer, primary_key=True)
    producer_name = db.Column(db.String(100), nullable=False)
    stocks = db.relationship('Stock', backref='supplier', lazy=True)

class Contact(db.Model):
    contact_id = db.Column(db.Integer, primary_key=True)
    message = db.Column(db.String(100), nullable=False)
    email = db.Column(db.String(255))
    phonenumber = db.Column(db.String(20))

```

On future fixes/changes 2 directly stand out to me, firstly is a filtering system in the admin dashboard that only shows the producers products corresponding to which administrator account is logged in. This is a must for integrity and overall usability (a producer doesn't want to see/alter stock they don't even produce.). the second change goes back to my database; the product tables remain empty no matter what happens in the web app.

|            |          |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                      |
|------------|----------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------|
| Table:     | product  |  |  |  |  |  |  |  |  |  | Filter in any column |
| product_id | stock_id | order_id                                                                            | product_quantity                                                                    | total_price                                                                         | discount_amount                                                                     | discount_price                                                                      |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                      |

Signalling prices would be nice aspect of the database. This is a great quality of life update to have.